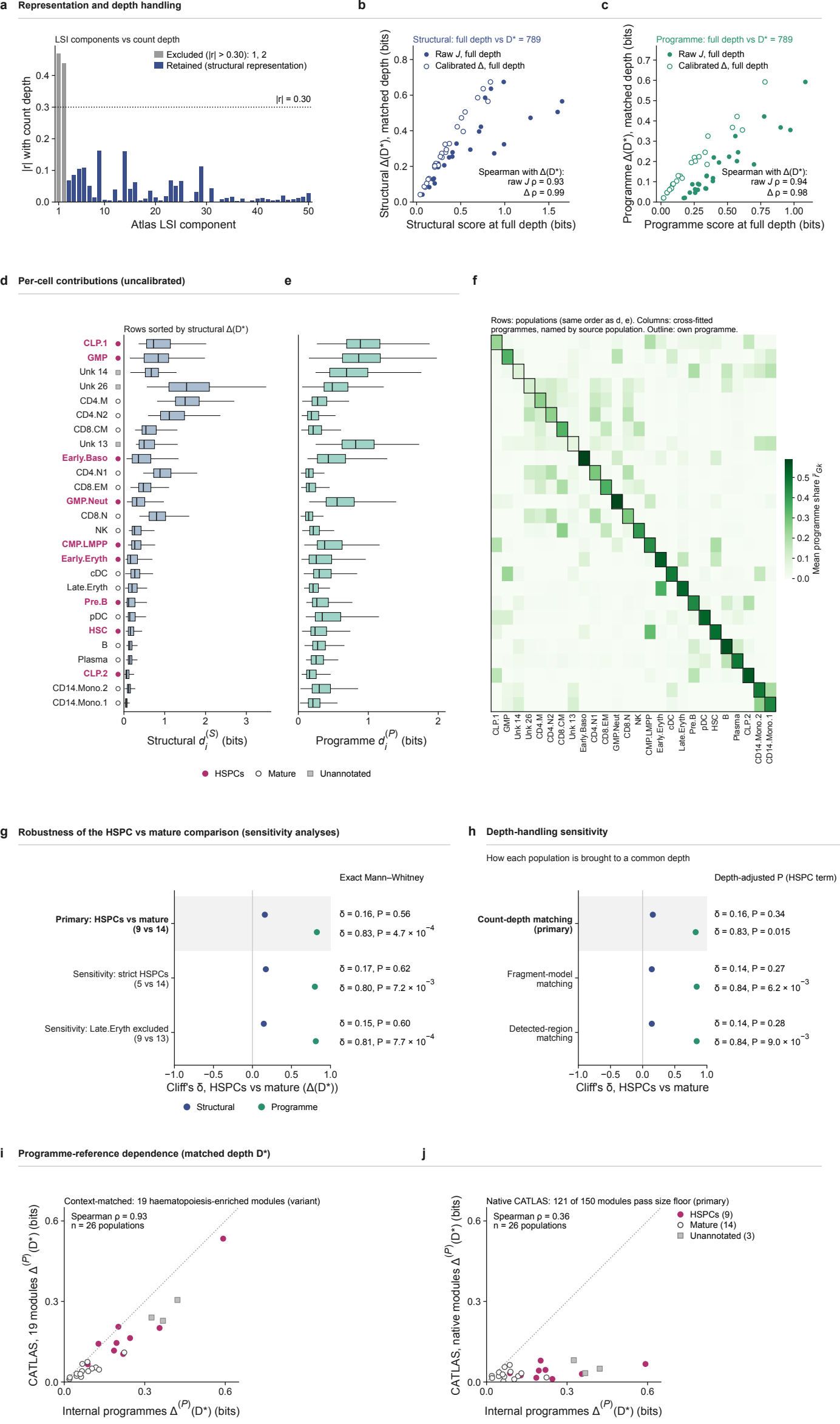

a
Representation and depth handling
LSI components vs count depth
Excluded (|r| > 0.30): 1, 2
Retained (structural representation)
|r| = 0.30
|r| with count depth
Atlas LSI component
b
Structural: full depth vs D* = 789
Raw J, full depth
Calibrated Δ, full depth
Spearman with Δ(D*):
raw J ρ = 0.93
Δ ρ = 0.99
Structural Δ(D*), matched depth (bits)
Structural score at full depth (bits)
c
Programme: full depth vs D* = 789
Raw J, full depth
Calibrated Δ, full depth
Spearman with Δ(D*):
raw J ρ = 0.94
Δ ρ = 0.98
Programme Δ(D*), matched depth (bits)
Programme score at full depth (bits)
d
Per-cell contributions (uncalibrated)
e
Rows sorted by structural Δ(D*)
CLP.1
GMP
Unk 14
Unk 26
CD4.M
CD4.N2
CD8.CM
Unk 13
Early.Baso
CD4.N1
CD8.EM
GMP.Neut
CD8.N
NK
CMP.LMPP
Early.Eryth
cDC
Late.Eryth
Pre.B
pDC
HSC
B
Plasma
CLP.2
CD14.Mono.2
CD14.Mono.1
Structural $d_i^{(S)}$ (bits)
Programme $d_i^{(P)}$ (bits)
HSPCs
Mature
Unannotated
f
Rows: populations (same order as d, e). Columns: cross-fitted programmes, named by source population. Outline: own programme.
Mean programme share $\bar{r}_{Gk}$
g
Robustness of the HSPC vs mature comparison (sensitivity analyses)
Exact Mann–Whitney
Primary: HSPCs vs mature (9 vs 14)
δ = 0.16, P = 0.56
δ = 0.83, P = 4.7 × 10^−4
Sensitivity: strict HSPCs (5 vs 14)
δ = 0.17, P = 0.62
δ = 0.80, P = 7.2 × 10^−3
Sensitivity: Late.Eryth excluded (9 vs 13)
δ = 0.15, P = 0.60
δ = 0.81, P = 7.7 × 10^−4
Cliff's δ, HSPCs vs mature (Δ(D*))
Structural
Programme
h
Depth-handling sensitivity
How each population is brought to a common depth
Depth-adjusted P (HSPC term)
Count-depth matching (primary)
δ = 0.16, P = 0.34
δ = 0.83, P = 0.015
Fragment-model matching
δ = 0.14, P = 0.27
δ = 0.84, P = 6.2 × 10^−3
Detected-region matching
δ = 0.14, P = 0.28
δ = 0.84, P = 9.0 × 10^−3
Cliff's δ, HSPCs vs mature
i
Programme-reference dependence (matched depth D*)
Context-matched: 19 haematopoiesis-enriched modules (variant)
Spearman ρ = 0.93
n = 26 populations
CATLAS, 19 modules Δ(P) (D*) (bits)
Internal programmes Δ(P) (D*) (bits)
j
Native CATLAS: 121 of 150 modules pass size floor (primary)
Spearman ρ = 0.36
n = 26 populations
HSPCs (9)
Mature (14)
Unannotated (3)
CATLAS, native modules Δ(P) (D*) (bits)
Internal programmes Δ(P) (D*) (bits)